SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I **Narappagari Sravya(192472192)(**Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:sravya

Annexure A

<u>Constraint-Based Problem Solving</u>

1. **A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:**
   Constraints:

   i.D1 cannot work Night shift
   ii.D2 must work before D3 (Morning < Afternoon < Night)
   iii.Only one doctor per shift
   iv.D3 cannot work Morning
   v.All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

   Problem

   Assign 3 doctors (D1, D2, D3) to 3 shifts:

   - Morning (M)
   - Afternoon (A)
   - Night (N)

   Constraints

   1. D1 cannot work Night.
   2. D2 must work before D3 (Morning < Afternoon < Night).
   3. Only one doctor per shift.
   4. D3 cannot work Morning.
   5. Every doctor gets exactly one shift.

State Space

Doctors = {D1, D2, D3}

Shifts = {Morning, Afternoon, Night}

Backtracking Algorithm

1. Start with an empty assignment.

2. Assign a shift to D1.

3. Check constraints.

4. Assign a shift to D2.

5. Check constraints.

6. Assign a shift to D3.

7. If all constraints are satisfied, return solution.

8. Otherwise backtrack.

Step-by-Step Backtracking

Step 1

Assign

D1 → Morning

Check

- D1 ≠ Night ✓

Assignment

D1 → Morning

Step 2

Try

D2 → Morning

Already occupied ❌

Reject.

Try

D2 → Afternoon

Available ✓

Assignment

D1 → Morning

D2 → Afternoon

Step 3

Remaining shift

Night

Assign

D3 → Night

Check constraints

- D3 ≠ Morning ✓
- D2 before D3

Afternoon < Night ✓

All shifts unique ✓

All doctors assigned ✓

Solution found.

Constraint Checking Table

Constraint	**Status**
D1 not Night	✓
D2 before D3	✓
One doctor per shift	✓
D3 not Morning	✓
Every doctor assigned	✓

Final Schedule

Doctor	**Shift**
D1	Morning
D2	Afternoon
D3	Night

2. **A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.**

Constraints:

i.Movement allowed: Up, Down, Left, Right
ii.Cost of each move = 1
iii.Cannot pass through obstacles
iv.Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

Assume the following 5 × 5 Grid

S . . X .

. X . X .

. X . . .

. . X X .

. . . . G

Legend

S = Start (0,0)

G = Goal (4,4)

X = Obstacle

Movement

- Up
- Down
- Left
- Right

Cost = 1

Heuristic (Manhattan Distance)

$$h(n) = | x_g - x | + | y_g - y |$$

Goal = (4,4)

Example

For Start (0,0)

$$h = | 4 - 0 | + | 4 - 0 | = 8$$

BFS Node Expansion

Initial Queue

[(0,0)]

Visited

(0,0)

Expand (0,0)

Children

(1,0)

(0,1)

Queue

[(1,0),(0,1)]

Expand (1,0)

Children

(2,0)

Queue

[(0,1),(2,0)]

Expand (0,1)

Children

(0,2)

Queue

[(2,0),(0,2)]

Expand (2,0)

Children

(3,0)

Queue

[(0,2),(3,0)]

Expand (0,2)

Children

(1,2)

Queue

[(3,0),(1,2)]

Continue expansion

$$\begin{array}{c}(3,0)\\ \downarrow\\ (3,1)\\ \downarrow\\ (4,1)\\ \downarrow\\ (4,2)\end{array}$$

↓

(4,3)

↓

(4,4)

Goal reached.

Optimal Path

(0,0)

↓

(1,0)

↓

(2,0)

↓

(3,0)

↓

(3,1)

↓

(4,1)

↓

(4,2

↓

(4,3

↓

(4,4)

Path Cost

Number of moves:8

Each move = 1

Total Cost = 8

Result

Optimal Path

S

↓

↓

↓

↓

→

↓

→

→

G

Total Cost:8

3. **An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.**

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- **The robot can move up, down, left, right**
- **Some cells are blocked (obstacles) and cannot be traversed**
- **The robot has limited sensing capability (can only observe adjacent cells)**
- **Each movement has a cost of 1, but entering risky zones has an additional cost of 2**
- **The robot must minimize total cost while ensuring safe navigation**
- **The robot must update its knowledge dynamically (online search scenario)**

Tasks:

a) **Analyze all the given constraints and explain how they affect the robot's decision-making process.**

b) **Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved. c) Demonstrate how the robot applies AI concepts such as:**

- **Intelligent agents**
- **Rationality**
- **Environment type**

c) **Justify why your chosen approach is the most suitable for solving this problem under the given constraints.**

(a) Constraint Analysis

Constraint 1

Robot moves only

- Up
- Down
- Left
- Right

Effect

Robot cannot move diagonally.

Constraint 2

Obstacles

Blocked cells cannot be crossed.

Effect

Robot must search for alternate routes.

Constraint 3

Limited sensing

Robot observes only neighboring cells.

Effect

Environment is partially observable.

Knowledge is updated during movement.

Constraint 4

Movement Cost

Normal cell

1

Risky cell

1 + 2 = 3

Effect

Robot prefers safer paths.

Constraint 5

Minimize Cost

Robot selects the least-cost route instead of simply the shortest route.

Constraint 6

Online Search

New obstacles may be discovered.

Effect

Robot replans dynamically while moving.

(b) Solution Using Uniform Cost Search + Online Search

Step 1

Start at S.

Step 2

Observe adjacent cells.

Step 3

Insert available cells into the priority queue.

Priority = Total Path Cost

Step 4

Expand the node with the smallest cumulative cost.

Step 5

If a risky cell is encountered,

Cost += 2

Step 6

If a new obstacle appears,

Remove that path.

Recompute the best route.

Step 7

Continue until Goal is reached.

Flow

Start

↓

Observe neighbors

↓

Choose lowest-cost path

↓

Move

↓

Update map

↓

Obstacle found?

↓

Yes

↓

Replan

↓

Continue

↓

Goal

(c) AI Concepts

1. Intelligent Agent

The rescue robot

- Perceives environment
- Makes decisions
- Acts autonomously

Agent Loop

Perception

↓

Reasoning

↓

Action

↓

Feedback

↓

Repeat

2. Rationality

A rational agent always chooses the action that minimizes total expected cost while maximizing safety.

Example

Instead of taking

Shortest path = 8 moves

Risk = High

Robot chooses

9 moves

Safe route

Lower overall risk.

3. Environment Type

Property	Type
Observability	Partially Observable
Agents	Single Agent
Dynamics	Dynamic
State Space	Discrete
Knowledge	Unknown initially

(d) Justification

The best approach is **Uniform Cost Search combined with an Online Search Agent** because:

- It always finds the lowest-cost path.
- It considers both movement and risk costs.
- It can adapt to newly discovered obstacles.
- It works effectively in partially observable environments.
- It supports dynamic replanning without requiring complete knowledge of the environment.

Final Results

Question 1

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Question 2

Optimal Path:

S → (1,0) → (2,0) → (3,0) → (3,1) → (4,1) → (4,2) → (4,3) → G

Total Cost = 8

Question 3

- Environment: Partially observable, dynamic, discrete.
- Search Strategy: Uniform Cost Search with Online Search.
- AI Concepts: Intelligent agent, rational decision-making, dynamic environment adaptation.
- Reason: Minimizes total cost, avoids risky areas, and replans when new information becomes available.